

# The Perfect Matching Problem

$$P_{\text{perfect-matching}}(G) = P_{\text{PM}}(G) \\ = \text{conv} \{x^M : M \text{ is a perfect matching}\},$$

A matching  $M$  is perfect if every  $v$  is in  $M$ .

$$\sum_{e \in \delta(v)} x_e \stackrel{\text{X}}{=} 1 \quad \forall v, \quad x_e \in \{0, 1\}.$$

$$Q_{\text{PM}}(G) = \left\{ x_e \geq 0, \underbrace{\sum_{e \in \delta(v)} x_e = 1}_{\forall v} \right\} \stackrel{?}{=} P_{\text{PM}}(G).$$

Thm:  $Q_{\text{PM}}(G) = P_{\text{PM}}(G).$

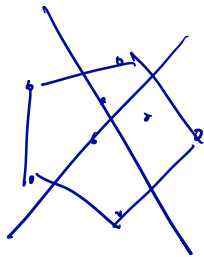
pf: Take  $x \in Q_{\text{PM}}(G) \subseteq Q_{\text{matching}}(G) = P_{\text{matching}}(G) = \overbrace{\sum \alpha_M x^M}$

Suppose  $\alpha_{\hat{M}} > 0$  for  $\hat{M} \notin P_{\text{PM}}$ . Let  $\hat{v}$  be a node not in  $\hat{M}$ .

$$\sum_{e \in \delta(\hat{v})} x_e = \sum_{M \text{ covering } \hat{v}} \alpha_M = 1 - \alpha_{\hat{M}} < 1 \quad *.$$

## Vertices are Perfect Matchings

Thm: Vertices of  $P_{PM}(G)$  and  $\{\chi_M : M \text{ a p.m.}\}$  are the same.



Hence a p.m.  $\chi_i^M$  cannot be expressed as convex comb. of other matchings.

pf: Straight forward,

## Birkhoff - von Neumann

A matrix  $X$  is called

Doubly stochastic if  $X_{ij} \geq 0$

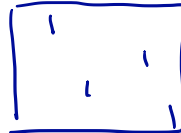
$$\text{and } \sum_i X_{ij} = 1 \quad \forall j$$
$$\sum_j X_{ij} = 1 \quad \forall i$$

Exercise

Fill in details  
(What is bipartite  
graph? etc.).

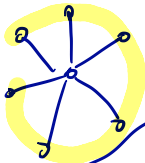
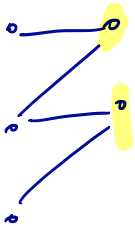
We showed:  $Q_{PM} = P_{PM}$

Says:  $X$  is DS iff it is a conv. comb.  
of permutation matrices.



## Minimum Vertex Cover / König

$G = (V, E)$ , a subset  $S \subseteq V$   
is called a vertex cover if  $\forall e \in E$   
 $e$  is incident to some vertex in  $S$ .



Thm: (König)  
For  $G = (V, E)$  bipartite,  
the cardinality of the max matching  
and min vertex cover, are equal.

# Proof of König — 1/2

Max Matching

$$\max: \sum x_e$$

$$\text{st: } x_e \geq 0$$

$$\sum_{e \in \delta(v)} x_e \leq 1 \quad \forall v \in V$$

~~$$x_e \in \{0, 1\}$$~~

min vertex cover  
 $\swarrow \leftarrow$

LP Relaxation

Min Vertex Cover

$$\min: \sum y_v$$

$$\rightarrow y_{v_1} + y_{v_2} \geq 1, \quad \forall (v_1, v_2) \in E$$

~~$$y_v \in \{0, 1\}$$~~

$$\underline{\underline{y_v \geq 0 \quad \forall v}}$$

Exercise: The LPs are duals of each other.

# Proof of König — 2/2

Induction on  $|V|$ . If  $|V| \leq 2$

Let  $y^*$  solve the LP relaxation for min vert. cover.  
i.e.,  $y_{v_1}^* + y_{v_2}^* \geq 1 \quad \forall (v_1, v_2) \in E, \quad y^* \geq 0.$

$E \neq \emptyset \Rightarrow y_{\hat{v}}^* > 0$  some  $\hat{v}$ .

CS  $\Rightarrow \sum_{e \in \delta(\hat{v})} x_e = 1$  holds for any opt. solution.

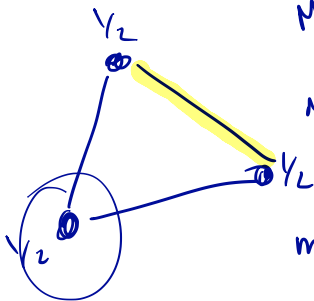
holds for all  $x^M$  when  $M$  is a maximal matching.

Every maximal matching covers  $\hat{v}$ .

$$\Rightarrow \max\_match(G - \hat{v}) = \max\_match(G) - 1.$$

$$\min\_vertex\_cov(G) \leq \max\_match(G - \hat{v}) + 1 = \max\_match(G) = \max\_match(G)$$

Reverse inequality follows from duality; relaxation.



Max Match : size 1

min vertex cover : size 2

$$\min: \sum y_v$$

$$\text{st: } y_{v_1} + y_{v_2} \geq 1 \quad \forall (v_1, v_2)$$

$$y_v \geq 0$$

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$$y^* = \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \end{pmatrix}$$